# FC7xxx Port Integration Manual

Rev A0

# Revision History

| Revision | Date | Changes |
|---|---|---|
| A0 | 2025/01/14 | Initial release |

# Table of Contents

# Chapter 1 Introduction

## 1.1 Introduction

This integration manual describes the integration requirements for the Port module.

# Chapter 2 Building

## 2.1 Dependencies on Other Modules

**Module configuration dependency**

- Mcu: This module provides the clock reference point for Port module.
- Common: This module is the basic module which used to choose the chip.

**Module build dependency**

- Common: This module provides common type for other modules.
- Rte: This module provides APIs to protect/unprotect some parts of code from interrupts (Exclusive Areas).
- Det: This module is necessary for enabling Development error detection.

**Module initialization dependency**

- Mcu: To use the Port module, the user needs to initialize the MCU module first.

## 2.2 Files Required for Compile

**Port module files:**

- _MCAL/Src/ Port /Src/ Port.c
- _MCAL/Src/Port/Src/Port_Hw.c
- _MCAL/Src/Port/include/ Port.h
- _MCAL/Src/Port/include/Port_Hw.h
- _MCAL/Src/Port/include/Port_Hw_Types.h
- _MCAL/Rte/Port/include/Port_MemMap.h
- _MCAL/Src/Port/include/Port_RegOps.h
- _MCAL/Src/Port/include/Port_Version.h
- _MCAL/Src/Port/include/Gpio_RegOps.h
- _MCAL_generate/src/Port_PBcfg.c
- _MCAL_generate/src/Port_Cfg.c
- _MCAL_generate/include/Port_Cfg.h

**Common module files:**

- _MCAL/Src/Common/include/Mcal.h
- _MCAL/Src/Common/include/Std_Types.h
- _MCAL/Src/Common/include/Platform_Types.h
- _MCAL/Src/Common/include/Compiler.h
- _MCAL/Src/Common/include/Compiler_Cfg.h
- _MCAL/Src/Common/include/CompilerDefinition.h
- _MCAL/Src/Common/include/Gpio_Reg.h
- _MCAL/Src/Common/ include/Port_Reg.h

**Det module files:**

- Det.h

**Rte module files:**

- SchM_Port.h

## 2.3 Add Plug-ins

Port module plug-ins are developed for EB tresos Studio, so, to use PORT plug-ins on the EB tresos Studio, the user needs to add the PORT module to the EB plug-ins folder first.

1) Copy the Port module(_MCAL/EB_Plugins/eclipse/plugins/Port) folder to EB tresos plug-ins (EB/tresos/plugins/) folder.
2) Set the Port module output location folder for the generated source file and header files.
3) Use EB tresos to configure the Port module and generate source and header files.

# Chapter 3 Memory

## 3.1 Sections in Memory Map

| Section Name | Section Type | Description |
|---|---|---|
| PORT_START_SEC_CONFIG_DATA_8<br>PORT_STOP_SEC_CONFIG_DATA_8<br>PORT_START_SEC_CONFIG_DATA_16<br>PORT_STOP_SEC_CONFIG_DATA_16<br>PORT_START_SEC_CONFIG_DATA_32<br>PORT_STOP_SEC_CONFIG_DATA_32 | Variables | These are all the sections used for variables which have to be aligned to 8/16/32 bit. These variables are initialized by startup code(data). |
| PORT_START_SEC_CONFIG_DATA_UNSPECIFIED<br>PORT_STOP_SEC_CONFIG_DATA_UNSPECIFIED | Configuration Data | Start and stop of Memory Section for Config Data. |
| PORT_START_SEC_CONST_BOOLEAN<br>PORT_STOP_SEC_CONST_BOOLEAN<br>PORT_START_SEC_CONST_8<br>PWM_STOP_SEC_CONST_8<br>PORT_START_SEC_CONST_16<br>PORT_STOP_SEC_CONST_16<br>PORT_START_SEC_CONST_32<br>PORT_STOP_SEC_CONST_32<br>PORT_START_SEC_CONST_ UNSPECIFIED<br>PORT_STOP_SEC_CONST_ UNSPECIFIED | Variables | These are all the sections used for variables which have to be aligned to 8/16/32 bit or boolean. These variables are read only (rodata). |
| PORT_START_SEC_CODE<br>PORT_STOP_SEC_CODE<br>PORT_START_SEC_ RAMCODE<br>PORT_STOP_SEC_ RAMCODE<br>PORT_START_SEC_CODE_AC<br>PORT_STOP_SEC_CODE_AC | Code | Start and stop of memory Section for Code(text). |
| PORT_START_SEC_VAR_NO_INIT_BOOLEAN<br>PORT_STOP_SEC_VAR_NO_INIT_BOOLEAN<br>PORT_START_SEC_VAR_NO_INIT_8<br>PORT_STOP_SEC_VAR_NO_INIT_8<br>PORT_START_SEC_VAR_NO_INIT_16<br>PORT_STOP_SEC_VAR_NO_INIT_16<br>PORT_START_SEC_VAR_NO_INIT_32<br>PORT_STOP_SEC_VAR_NO_INIT_32<br>PORT_START_SEC_VAR_NO_INIT_ UNSPECIFIED<br>PORT_STOP_SEC_VAR_NO_INIT_ UNSPECIFIED | Variables | These are all the sections used for variables which have to be aligned to 8/16/32 bit. These variables are never cleared and never initialized by startup code (bss). |
| PORT_START_SEC_VAR_ INIT_BOOLEAN<br>PORT_STOP_SEC_VAR_ INIT_BOOLEAN<br>PORT_START_SEC_VAR_ INIT_8<br>PORT_STOP_SEC_VAR_ INIT_8<br>PORT_START_SEC_VAR _INIT_16<br>PORT_STOP_SEC_VAR _INIT_16<br>PORT_START_SEC_VAR _INIT_32<br>PORT_STOP_SEC_VAR _INIT_32 | Variables | These are all the sections used for variables which have to be aligned to 8/16/32 bit. These variables are initialized by startup code (data). |

| Section Name | Section Type | Description |
|---|---|---|
| PORT_START_SEC_VAR _INIT_ UNSPECIFIED<br>PORT_STOP_SEC_VAR _INIT_ UNSPECIFIED | | |
| PORT_START_SEC_VAR_INIT_SHAREABLE_NO_CACHEABLE<br>PORT_STOP_SEC_VAR_INIT_SHAREABLE_NO_CACHEABLE | Variables | These are all the sections used for variables which have to be aligned to 8/16/32 bit. These variables are initialized by startup code (share). |

# Chapter 4 Exclusive Area

Port module using the services of Schedule Manger (SchM) for entering and exiting critical regions.

The following critical regions are used in the Port driver:

- Port.c:
  - Port_Init : exclusive area 5, exclusive area 6
- Port_Hw.c:
  - Port_LL_Init_Port : exclusive area 0
  - Port_LL_SetPinDirection : exclusive area 1
  - Port_LL_SetPinMode : exclusive area 2
  - Port_LL_SetGpioPinOutput : exclusive area 3
  - Port_LL_RefreshPortDirection : exclusive area 4
  - Port_LL_GetLock: exclusive area 7
  - Port_LL_ReleaseLock: exclusive area 7

# Chapter 5 Interrupt Service Routine (ISR)

None.

# Chapter 6 Error Report

## 6.1 Det

| Function Name | Error Type |
| --- | --- |
| Port_Init | PORT_E_INIT_FAILED<br>PORT_E_GET_SPIN_LOCK_FAILED |
| Port_SetPinDirection | PORT_E_UNINIT<br>PORT_E_PARAM_PIN<br>PORT_E_DIRECTION_UNCHANGEABLE<br>PORT_E_GET_SPIN_LOCK_FAILED |
| Port_SetPinMode | PORT_E_UNINIT<br>PORT_E_PARAM_PIN<br>PORT_E_PARAM_INVALID_MODE<br>PORT_E_MODE_UNCHANGEABLE<br>PORT_E_GET_SPIN_LOCK_FAILED |
| Port_RefreshPortDirection | PORT_E_UNINIT<br>PORT_E_GET_SPIN_LOCK_FAILED |
| Port_GetVersionInfo | PORT_E_PARAM_POINTER |

## 6.2 Dem

None

# Chapter 7 Function Calls

## 7.1 Function Calls during Startup

None

## 7.2 Function Calls during Shutdown

None

## 7.3 Function Calls during Wake-up

None

## 7.4 Function Calls during Runtime

None

# Chapter 8 Other Requirements

## 8.1 Notification, Callback, Callout

None

## 8.2 Macros

None

# Chapter 9 Integration Steps

1) Configure the Port module and generate configuration files (please refer to Building chapter for details).
2) Configure appropriate memory sections in linker file or other (please refer to Memory chapter for details).
3) Compile and build the Port with all the dependent modules.